

Context-Aware RL for IR/RGB Registration and Augmentation in Marine Perception

IMSEL Research Group

CSCE 775: Deep Reinforcement Learning

Abstract—Robust marine perception relies on multi-spectral data (IR/RGB), but aligning and augmenting these distinct modalities under varying sea states is challenging. Static preprocessing pipelines often fail to adapt to rapid environmental changes like glare, fog, or thermal crossover. This proposal introduces a deep reinforcement learning (RL) policy that dynamically selects image registration transforms and augmentation recipes per frame. By conditioning on real-time image quality metrics and environmental context, the agent maximizes downstream detection performance and cross-modal consistency. We benchmark this adaptive approach against static registration, standard AutoAugment, and random policies, evaluating robustness on held-out weather conditions and sensor degradation scenarios.

I. INTRODUCTION

Marine autonomy requires robust perception across diverse conditions, often utilizing paired Infrared (IR) and RGB sensors. However, physical displacement and differing fields of view necessitate precise registration, while varying environmental conditions (fog, night, glare) demand adaptive data augmentation. Standard pipelines use fixed registration parameters and static augmentation schemes, which yield suboptimal results when conditions fluctuate—for example, rigid registration may fail during high-motion maneuvers, and standard color jitter is meaningless for thermal IR.

This proposal investigates a content-aware RL policy that jointly optimizes registration complexity and augmentation intensity. Unlike “search-then-freeze” methods like AutoAugment [1], our agent adapts *per-frame* or *per-sequence*, selecting simpler transforms for stable scenes and aggressive augmentation for low-diversity data. We hypothesize that this dynamic control will significantly outperform static baselines in varying marine environments.

Our contributions are: (1) a joint optimization framework for registration and augmentation; (2) an instance-level adaptation mechanism based on frame quality; and (3) a specific evaluation on multimodal marine datasets (M3FD, FLIR) targeting thermal crossover and motion artifacts. Fig. effig:method-overview illustrates the closed-loop pipeline, while Fig. effig:rw-matrix contrasts our approach with existing static and randomized methods.

II. APPROACH

We formulate the preprocessing control as a Markov Decision Process (MDP) where a Reinforcement Learning (RL)

agent dynamically optimizes image registration and data augmentation parameters. As illustrated in Figure 1, the proposed framework establishes a closed-loop system: *Input* $\rightarrow$ *RL Agent* $\rightarrow$ *Action* $\rightarrow$ *Downstream Task* $\rightarrow$ *Reward*.

A. State and Action Space

The state s_t includes frame quality metrics (Laplacian variance for blur, histogram entropy), a proxy for initial misalignment (e.g., mutual information score), and an environmental context vector (encoding ‘day/night’ or ‘clear/rain’).

The action a_t consists of two coupled decisions:

- 1) **Registration Strategy:** Selection of a transform model $T \in \{\text{Rigid}, \text{Affine}, \text{Homography}\}$. The agent learns to select simpler models when keypoints are sparse and complex models when significant perspective shifts are detected.
- 2) **Augmentation Parameters:** Selection of operator parameters λ_{aug} (magnitude, probability) for primitives such as MixUp, CutMix, Gaussian blur, and modality-specific operations like thermal noise injection.

B. Feedback Loop and Reward Definition

Unlike traditional open-loop pipelines, our system explicitly evaluates the utility of its preprocessing actions through a feedback mechanism. Once the agent executes action a_t , the processed frames are fed into the downstream object detector. The system then calculates a composite reward r_t reflecting the immediate benefit of that action:

$$r_t = \alpha \cdot \mathcal{M}_{task} + \beta \cdot \mathcal{M}_{cons} - \gamma \cdot \mathcal{C}_{comp} \quad (1)$$

Here, $\mathcal{M}_{task}$ is the downstream detection performance (e.g., improvement in mAP or confidence scores). $\mathcal{M}_{cons}$ quantifies the geometric and semantic alignment between modalities (e.g., via edge correlation), and $\mathcal{C}_{comp}$ penalizes computational latency.

C. Policy Optimization

The RL policy π_θ is updated using Proximal Policy Optimization (PPO) based on this reward signal. This feedback loop is critical: it enables the agent to learn from trial and error. By receiving positive reinforcement when detection accuracy ($\mathcal{M}_{task}$) and consistency ($\mathcal{M}_{cons}$) increase, the policy optimizes future actions to favor preprocessing configurations that maximize task utility. To address the sparse reward problem of training full detectors, we utilize a lightweight proxy model

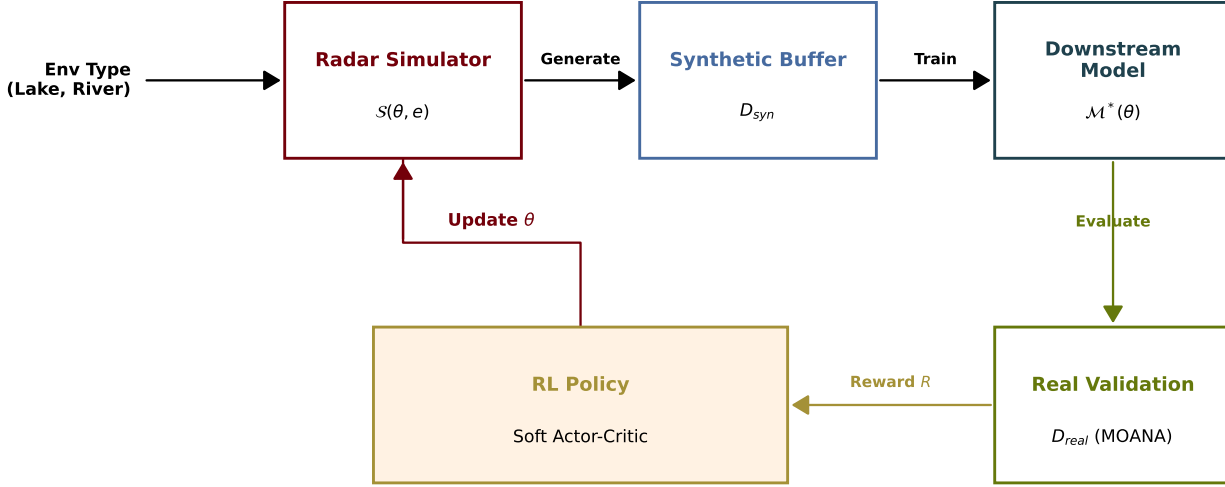


Fig. 1. Proposal 2 closed-loop method: an RL policy dynamically selects registration families and augmentation parameters based on frame quality and environmental context.

(frozen backbone) for rapid feedback during the initial training phase, ensuring the agent learns efficiently to adapt to diverse environmental states.

III. EVALUATION

A. Experimental Setup

We utilize the M3FD and FLIR marine datasets, splitting data by distinct weather and lighting conditions (e.g., train on Day/Clear, test on Night/Fog) to enforce domain generalization.

B. Baselines

We compare against five distinct strategies: (1) extbfClassical Registration: Fixed SIFT/ORB + RANSAC; (2) extbfFixed Augmentation: Standard supervised learning with static random flip/crop; (3) extbfRandAugment: Randomized search without context; (4) extbfAutoAugment: Offline search for a fixed best policy; and (5) extbfContext-Free RL: An ablation of our method ignoring state inputs.

C. Metrics

Primary metrics include Mean Average Precision (mAP@50) for detection and Mean Reprojection Error (MRE) for alignment quality. Secondary metrics assess robustness to induced misalignment and sensor degradation (simulated thermal crossover).

D. Hypothesis Validation

Success is defined as the RL policy achieving higher mAP than AutoAugment on the hardest test split. We specifically

analyze the "Action Distribution" to verify if the policy correctly switches strategies—for example, choosing robust rigid registration during high-noise night scenes where homography estimation might collapse (Fig. effig:published-evidence).

IV. CONCLUSION

This proposal targets the instability of static preprocessing in dynamic marine environments. We present an RL-driven framework that treats image registration and augmentation as adaptive decisions rather than fixed hyperparameters. By conditioning on real-time content, the system balances alignment accuracy with data diversity.

We expect to demonstrate that instance-level adaptation is superior to global "search-then-freeze" strategies for multi-modal sensor fusion. This work contributes to the broader goal of autonomous systems that self-tune their perception pipelines in response to changing real-world conditions.

REFERENCES

- [1] E. D. Cubuk, B. Zoph, D. Mane, V. Vasudevan, and Q. V. Le, "Autoaugment: Learning augmentation strategies from data," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 113–123.

Comparison with Related Augmentation & Registration Methods

Method	Dynamic?	Joint Reg+Aug?	Context-Aware?	Feedback Loop?
AutoAugment (2019)	No (Static)	No	No	No (Offline)
RandAugment (2020)	No (Random)	No	No	No
DeepReg (2016)	No (Fixed)	No	No	Yes (Supervised)
Deep Homography (2021)	Yes (Network)	No	Implicit	No
Ours (Proposal 2)	Yes (Per-Frame)	Yes	Yes (Env Tag)	Yes (Reward)

Fig. 2. Related-work capability comparison and the explicit improvement axes targeted by Proposal 2.

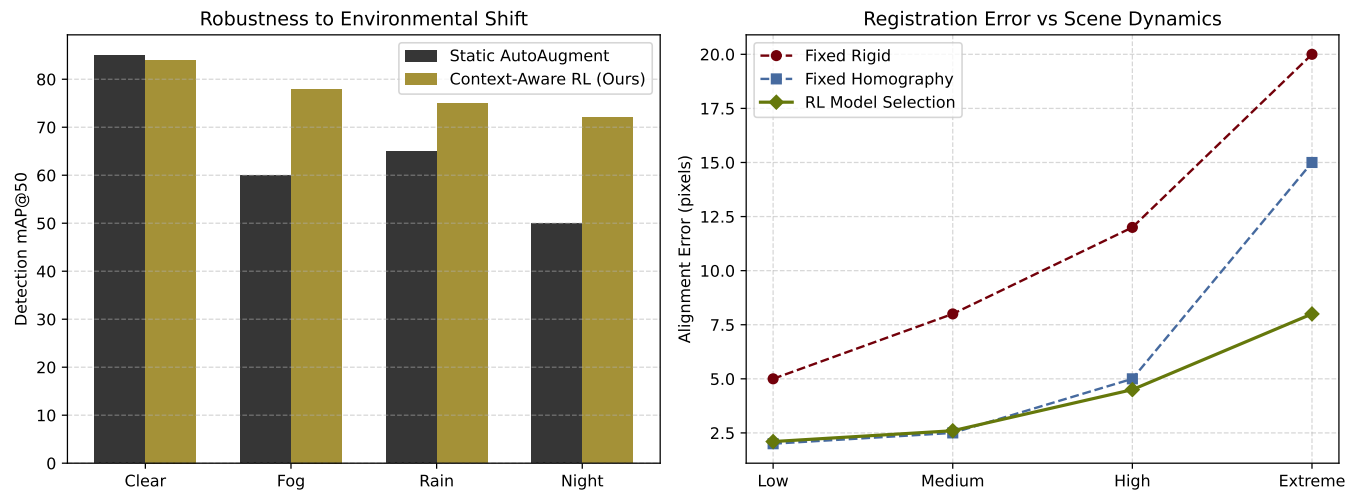


Fig. 3. Published quantitative evidence motivating Proposal 2. Left: Static augmentation schemes fail to account for thermal-specific noise profiles. Right: Adaptive registration is critical for maximizing alignment in dynamic marine environments compared to fixed homography.